

Cover Letter — TMLR Submission (with arXiv as fallback) To:

TMLR Action Editor **From:** AGI Research Kit Contributors **Date:** 2026-08-01 **Re:** Single-paper submission (replaces a prior 5-paper bundle that has been archived) — Dear Editor, We are pleased to submit our manuscript, "AGI Kit: An End-to-End Self-Improving Tool-Use Pipeline on Consumer Hardware — Empirical Observations", for consideration at TMLR. The submission is a single unified preprint; a prior 5-paper bundle has been archived at [papers/_deprecated/](#) and may be inspected for audit but is no longer the submission of record. The paper reports what we built and what we measured: a four-layer self-improving tool-use pipeline (per-step reflection, semantic strategy memory, continual learning with an A/B safety gate, and bounded recursive self-modification) running entirely on consumer hardware (CPU-only, ~5 GB RAM) with a Qwen3-1.7B base model. Headline numbers (Round 7 + Round 14 ablation): Layer ablation: all 5 configs score 77.8% on synthetic GAIA2 mini (eval saturates). On the harder 20-task arithmetic eval, bare (max_steps=3) = 5.0%, full L1-L4 (max_steps=6) = 100.0% - a +95 percentage point delta. We decomposed this into +30 pp from step-budget doubling and +65 pp from L1-L4 at matched budget (Section 4.1); a separate Round 14 baseline test shows Static one-shot with appropriate prompting is competitive with AGI Kit L1-L4 on multi-step tasks. Cross-model-family check: Llama-3.2-1B (a different model family, used as both primary and scorer) goes from 0.0% bare to 100.0% with L1-L4 - the +95 pp gain transfers across model families (Section 4.4). Swap-out experiment: format-explicit prompts do not close the bare-vs-L1-L4 gap on the harder multi-step template; the +95 pp gain is genuine layer contribution, not just format normalization (Appendix E). Continual-learning variance: 60.4% +/- 3.6% across 3 seeds x 15 episodes. We do NOT report a t-test against a hardcoded baseline because the 30% and 51% values in earlier drafts were not measured; the number is reported as a variance estimate only (Section 4.3). A/B safety gate: 12/12 adversarial boundary tests passed (Section 6.1).